

Tarun Mittal

Senior Software Engineer

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SUMMARY

- Senior Software Engineer with **5 years of experience** designing and owning cloud-native distributed systems.
- Expertise in **Python microservices**, OLAP data platforms (**ClickHouse**, **PostgreSQL**), event-driven systems, and **Agentic AI and RAG**.
- Led backend migrations end to end and **mentored engineers** at Uber, Masters India, and Impact Analytics.

SKILLS

- **Languages:** Python, Go (Golang), SQL.
- **Backend & APIs:** FastAPI, Gin, Django, SQLAlchemy, Pydantic, REST, gRPC, Celery, asyncio.
- **AI & Agents:** LLM agents (OpenAI/Gemini/Claude), LangGraph, LangChain, MCP, RAG, pgvector.
- **Data & Streaming:** Kafka, Flink, Spark, Pinot, ClickHouse, BigQuery, PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, S3, CDC, CQRS.
- **Cloud & Infra:** GCP, AWS, Docker, Kubernetes, CI/CD, Bazel, Observability (ELK, New Relic, Prometheus, Grafana).
- **Core:** Distributed Systems, Microservices, System Design (HLD/LLD), Concurrency, High Availability, Caching, Mentorship, Unit Testing.

PROFESSIONAL EXPERIENCE

Senior Software Engineer

IMPACT ANALYTICS, Bangalore

May 2026 – Present

Agentic AssortSmart Platform & PostgreSQL to ClickHouse Data Migration

- Building the **Cluster Recommendation Copilot** for AssortSmart, an agentic store clustering module where an LLM agent (**LangGraph**, **MCP**) plans and calls **14 audited tools** while a deterministic engine computes clusters, with human approval gates before any write. Targets are plan turnaround **under 1 hour** (from days), **20+ configurations** evaluated per run (from 1), and run failures **under 2%** (from a measured 8.5%).
- Benchmarked **PostgreSQL vs ClickHouse** on production retail workloads. The Order Batching metric over **23.7M joined rows** ran in **3.86s on ClickHouse vs 3m 40s+ on PostgreSQL (60x)**, parallel bulk loads sustained **5.9M rows/sec vs 250K rows/sec**, and query to CSV export throughput improved **43x** (607 to 26,400 rows/sec).
- Designed the Order Batching **CQRS read path migration** keeping writes on PostgreSQL and serving analytical reads from ClickHouse, with **CDC mirrors** for hot fact tables, daily refresh for low churn dimensions, and **Redis read your writes routing (30s TTL)** so users see their own saves instantly.
- Cut 10K row update handling on a **29M row fact table from 39s to 7s** using partition scoped merges over full table rewrites, and validated **ReplacingMergeTree** versioned updates, partition pruning, and dictionary lookups (17M row fact table compressed **7.8x** to 433 MiB).
- **Tech Used:** Python, FastAPI, LangGraph, MCP, ClickHouse, PostgreSQL, CDC, Redis, Kafka, GCP, Docker.

Software Development Engineer 2

UBER (via EPAM Systems), Bangalore

July 2024 – May 2026

Financial Risk Management (FRM) Scoping Platform (Uber Finance)

- Built Uber's quarterly **FRM Risk Scoping platform (FastAPI, MySQL, React)** replacing a manual Google Sheets workflow across **8 screens and 30+ REST endpoints**, with scoping output feeding **PwC audit work papers at \$340M group materiality**, targeting a **70% cut** in manual ingestion and reconciliation effort per close.
- Designed the **11 table SQLAlchemy 2.0 MySQL schema** (balance sheet, income statement, EMI, thresholds, component and residual scoping, recon) behind a layered **handler, service, repository** architecture with header based auth middleware.
- Owned the migration of reconciliation from Sheets backed v1 APIs to **MySQL backed v2 APIs**, validating HFM vs public **10-Q** amounts, and fixed a latent **column aliasing bug** surfaced during the ORM move.
- **Led a pod of 3 engineers**; grew the service to **1,100+ pytest unit tests** under Bazel with mocked handler and repository suites gating CI.
- **Tech Used:** Python, FastAPI, SQLAlchemy 2.0, Pydantic, MySQL, Bazel, pytest, Docker.

Menu Ingestion & Automation Platform (Uber Eats)

- Built an event-driven pipeline with **Kafka (200 to 500 peak events/sec)**, **Flink**, and **Spark** for **30,000+ vendor menus/month**, cutting onboarding from 24h to 2h and saving **\$600K+ annually** by eliminating a \$2 per menu third party tool.
- Built an AI extraction pipeline with **RAG and Gemini** converting PDF and image menus into schema validated catalog data with **98% fidelity and 100% schema consistency** via supervised fine tuning.
- Increased successful ingestions by **95%** (roughly 60% to 95%+) via IP rotation and proxy pools; built ingestion health dashboards on **Apache Pinot** with sub-second queries, cutting failure detection from hours to minutes.
- **Tech Used:** Python, Apache Kafka, Apache Flink, Apache Spark, Apache Pinot, Selenium, Gemini, GCP, Docker.

Software Development Engineer 2

MASTERS INDIA, Noida

December 2022 – June 2024

GST Compliance & E-Invoicing Platform

- Led the migration from a PHP (Laravel) monolith to asynchronous **Python (FastAPI) microservices**, **mentoring 2 engineers** through the rollout and cutting p95 API latency from **1.2s to 300ms** for **1,500+ enterprise clients**.
- Scaled bulk e-invoice processing to **100,000+ transactions per import and 1M+ daily transactions** (12 TPS average, 100+ TPS at filing deadline peaks) with async workers and idempotent retries, raising sustained throughput from **700 to 4,000 requests/min**.

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- Added **Redis caching** that cut redundant database reads by **30%**, and shipped an **audit logging** feature that reduced client churn by **15%**.
- Set up **ELK and New Relic alerting**, cutting incident triage by **70%** (about 30 min to under 10), and raised test coverage from **35% to 82%** with pytest gating CI/CD at a **98% deployment success rate**.
- **Tech Used:** Python, FastAPI, PostgreSQL, MongoDB, Redis, Elasticsearch, Celery, Docker, ELK, New Relic, AWS.

Software Development Engineer

GEEKSFORGEEKS, Noida

August 2021 – November 2022

- Redesigned and migrated the backend from PHP to **Django (Python)**, improving reliability for **10,000+ daily queries**.
- Designed data models and REST APIs for voting and pinning, lifting **premium subscriptions 15 to 20%** (Django, MySQL, MongoDB, Redis, Elasticsearch, AWS).
- Built an influencer dashboard with real time analytics and cron pipelines for video processing and reminders, boosting **course sales 30%** and operational efficiency **70%**.

EDUCATION

B.Tech (Information Technology), CGPA 7.7/10

Institute of Engineering & Technology,
Lucknow

July 2017 – June 2021

ACHIEVEMENTS

- Ranked **2260 / 37,000+** in the Google Code Jam Qualifier Round 2021.
- Finalist (top 3 nationally) at the **Smart India Hackathon 2020 (MHRD)**.
- Participated in the **Global AI Hackathon** at EPAM Systems.